

TAK, Thailand

WMO: 483760

Lat: 16.8797N

Lon: 99.1400E

Elev: 127

StdP: 99.81

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	14.7	16.4	9.0	7.2	28.8	10.6	8.1	27.5	4.1	28.4	3.4	28.2	0.0	270	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.9	39.6	24.2	38.5	23.9	37.4	23.8	27.3	32.8	26.9	32.5	26.6	32.0	1.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	21.8	29.6	25.6	21.2	29.2	25.2	20.7	28.8	87.4	33.2	85.6	32.5	84.1	32.3	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.5	4.9	4.2		12.1	41.5	1.7	1.1	10.9	42.3	9.9	42.9	9.0	43.5	7.7	44.3	
			WB	11.2	28.5	1.7	1.2	10.0	29.4	9.0	30.0	8.1	30.7	6.8	31.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	24.9	27.6	30.5	32.2	30.5	29.4	28.7	28.6	28.6	27.8	26.5	24.6
	DBStd	2.90	2.26	2.20	2.50	2.29	2.45	1.54	1.39	1.27	1.27	1.32	1.80	2.26
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6687	462	492	634	667	636	582	581	577	558	551	493	454
	CDD18.3	3646	204	259	376	417	378	332	322	318	308	293	243	196
	CDH23.3	42413	1994	3121	5154	5987	4843	3917	3633	3529	3364	2932	2239	1701
	CDH26.7	20196	890	1693	3039	3728	2561	1669	1366	1279	1293	1110	901	667
(14) Wind	WSAvg	1.0	0.5	0.7	1.1	1.4	1.4	1.4	1.6	1.6	1.0	0.4	0.3	0.4
(15) Precipitation	PrecAvg	1081	9	1	22	37	179	98	92	129	235	217	49	8
	PrecMax	1373	93	4	60	95	310	183	226	316	438	475	195	64
	PrecMin	765	0	0	0	1	46	27	44	46	92	60	0	0
	PrecStd	171	23	2	23	27	73	48	49	68	101	113	54	17
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	38.2	40.1	41.5	40.5	36.8	35.0	35.0	35.0	34.5	35.1	34.2
		MCWB	21.7	21.6	23.0	24.5	25.2	25.8	25.9	25.8	26.3	26.2	24.5	23.5
	2%	DB	34.0	36.9	38.9	40.1	38.4	35.0	33.7	33.5	33.9	33.4	33.6	32.9
		MCWB	21.8	21.9	22.8	24.5	25.6	26.0	25.7	25.7	26.1	26.0	24.7	23.3
	5%	DB	32.9	35.8	37.9	39.0	36.8	33.7	32.7	32.5	32.9	32.6	32.4	31.8
		MCWB	21.8	22.0	22.6	24.6	25.7	25.9	25.5	25.5	26.0	25.8	24.5	22.9
	10%	DB	31.6	34.6	36.8	37.8	35.1	32.6	31.6	31.4	31.9	31.7	31.3	30.4
		MCWB	21.6	22.1	22.6	24.6	25.5	25.7	25.3	25.3	25.8	25.6	24.1	22.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	25.7	26.7	27.4	27.7	27.6	27.2	27.2	27.7	27.4	26.7	25.8
		MCDB	30.3	33.9	35.2	37.3	34.3	32.3	32.0	31.7	32.5	31.8	32.0	31.5
	2%	WB	24.1	24.8	25.5	26.5	27.0	27.0	26.6	26.6	27.1	26.8	26.0	24.7
		MCDB	29.7	32.6	33.8	35.8	33.6	32.0	31.4	31.2	31.8	31.4	31.0	30.1
	5%	WB	23.4	24.1	24.8	26.0	26.6	26.6	26.1	26.2	26.6	26.4	25.5	24.1
		MCDB	29.2	31.6	33.3	35.1	33.0	31.4	30.8	30.6	31.1	30.9	30.0	29.2
	10%	WB	22.6	23.3	24.2	25.6	26.1	26.1	25.7	25.7	26.1	26.1	25.0	23.4
		MCDB	28.8	31.0	32.6	34.3	32.4	30.9	30.2	30.0	30.5	30.3	29.2	28.3
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	14.2	12.0	10.9	8.0	6.4	5.8	5.9	6.8	7.6	9.7	11.8
		MCDBR	15.3	15.4	13.3	11.9	10.1	8.0	7.1	7.3	8.1	9.0	11.2	13.0
		MCWBR	5.2	4.7	4.0	3.6	2.9	2.7	2.6	2.7	2.9	3.0	3.7	4.5
	5% WB	MCDBR	11.7	13.1	11.6	10.8	8.5	6.9	6.5	6.6	7.6	8.0	9.1	10.7
		MCWBR	4.4	5.0	4.2	3.6	3.0	2.8	2.9	2.9	3.0	3.1	3.3	4.1
(40) Clear-Sky Solar Irradiance	taub		0.465	0.555	0.672	0.619	0.492	0.436	0.426	0.432	0.463	0.538	0.467	0.463
	taud		2.134	1.910	1.659	1.829	2.222	2.403	2.428	2.400	2.295	2.035	2.212	2.160
	Ebn at Noon		817	760	687	728	815	854	864	867	842	765	811	806
	Edh at Noon		148	194	256	217	145	119	117	121	134	169	136	141
(44) All-Sky Solar Radiation	RadAvg		4.67	5.26	5.51	5.96	5.36	4.65	4.33	4.28	4.58	4.62	4.73	4.49
	RadStd		0.22	0.21	0.35	0.34	0.46	0.42	0.37	0.29	0.26	0.27	0.28	0.19

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.16	N/A	N/A	N/A	N/A

Nomenclature: See separate page